

INCIDENT INVESTIGATION SUMMARY

Seal Failure and Hydrocarbon Release, Pump P-101A

Incident No.: INC/2023/014
Unit: CDU-I
Classification: Process safety event, Tier 2

Date of Incident: 14 Aug 2023
Equipment: P-101A, crude charge pump
Investigation Lead: K. Menon, Reliability Engineer

1. Summary of Events

At approximately 0340 hours on 14 August 2023, the mechanical seal of crude charge pump P-101A failed, releasing an estimated 40 litres of hot crude to the pump plinth. The release did not ignite. The standby pump P-101B was started and the affected pump isolated within eleven minutes.

2. Causes

The immediate cause was cracking of the mechanical seal faces following a period of dry running. The seal flush line had become blocked with coke fines, interrupting the flush supply to the seal chamber.

The underlying cause was the absence of a low flow alarm on the seal flush line, and a plan review that had not identified the flush line as a safety critical element.

3. Timeline

Time	Event	Source
0318	Seal flush flow begins to decline	DCS trend
0334	Pump outboard bearing temperature rises 12 deg C	DCS trend
0340	Seal failure, hydrocarbon release to plinth	Field operator
0344	Gas detector LD-1104 in alarm	F&G log
0351	P-101B started, P-101A isolated	Shift log
0412	Area declared safe, clean up commenced	Shift log

4. Actions

S.No	Action	Owner	Target Date	Status
1	Install low flow alarm on seal flush lines of all crude charge p	Instrumentation	31 Dec 2023	Complete
2	Revise seal flush strainer cleaning frequency to monthly	Maintenance	30 Sep 2023	Complete
3	Add seal flush line to the safety critical element register	Reliability	31 Oct 2023	Complete
4	Include dry running scenario in operator competency refresh	Training	31 Mar 2024	In progress

Investigated By

Reviewed By

Approved By